
VECTOR SPACE PROBLEM SHEET

This sheet reinforces the lectures on vector spaces and inner products. Question 1 uses actual numbers; the later questions guide you, one step at a time, through proving some well known results. The algebra in each step is short. Try to explain each step in words as well as symbols: a good proof is one that someone else can follow without having to fill in gaps.

This looks very mathematical, but it helps to develop the tools and language that are used to describe machine learning.

Throughout, an inner product $\langle x, y \rangle$ between vectors in a vector space $\mathcal{V}$ satisfies the following properties

1. $\langle x, x \rangle \geq 0$ for all $x \in \mathcal{V}$
2. $\langle x, x \rangle = 0$ if and only if $x = 0$
3. $\langle \alpha x, y \rangle = \alpha \langle x, y \rangle$
4. $\langle x, y + z \rangle = \langle x, y \rangle + \langle x, z \rangle$
5. $\langle x, y \rangle = \langle y, x \rangle$

For vectors in $\mathbb{R}^n$ the usual inner product is $\langle x, y \rangle = x^\top y = \sum_{i=1}^n x_i y_i$, with norm $\|x\| = \sqrt{\langle x, x \rangle}$.

• Do not write in this space •

1 This question is a warm-up using actual numbers. Let

$$\mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}$$

with the usual inner product $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\top \mathbf{y}$ on $\mathbb{R}^3$.

(a) Compute $\langle \mathbf{x}, \mathbf{y} \rangle$, the norms $\|\mathbf{x}\|$ and $\|\mathbf{y}\|$, and the distance $d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$.

$$\begin{aligned} \langle \mathbf{x}, \mathbf{y} \rangle &= 1 \times 2 + 2 \times 0 + 2 \times 1 = 4, \\ \|\mathbf{x}\| &= \sqrt{1 + 4 + 4} = 3, \quad \|\mathbf{y}\| = \sqrt{4 + 0 + 1} = \sqrt{5}. \end{aligned}$$

$$\text{As } \mathbf{x} - \mathbf{y} = (-1, 2, 1)^\top, d(\mathbf{x}, \mathbf{y}) = \sqrt{1 + 4 + 1} = \sqrt{6} \approx 2.45.$$

[4 marks]

(b) Using $\langle \mathbf{x}, \mathbf{y} \rangle = \|\mathbf{x}\| \|\mathbf{y}\| \cos(\theta)$, find the angle θ between $\mathbf{x}$ and $\mathbf{y}$. Check that the Cauchy-Schwarz inequality, $|\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\| \|\mathbf{y}\|$, and the triangular inequality, $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$, both hold for these two vectors.

$$\cos(\theta) = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{x}\| \|\mathbf{y}\|} = \frac{4}{3\sqrt{5}} \approx 0.596 \Rightarrow \theta \approx 53.4^\circ.$$

Cauchy-Schwarz: $|\langle \mathbf{x}, \mathbf{y} \rangle| = 4 \leq 3\sqrt{5} \approx 6.71$. (Cauchy-Schwarz is exactly what guarantees $|\cos(\theta)| \leq 1$, so the angle always makes sense.) For the triangular inequality, $\mathbf{x} + \mathbf{y} = (3, 2, 3)^\top$ so $\|\mathbf{x} + \mathbf{y}\| = \sqrt{22} \approx 4.69 \leq 3 + \sqrt{5} \approx 5.24$.

[5 marks]

(c) The projection of $\mathbf{y}$ onto $\mathbf{x}$ is

$$\text{proj}_{\mathbf{x}}(\mathbf{y}) = \frac{\langle \mathbf{y}, \mathbf{x} \rangle}{\langle \mathbf{x}, \mathbf{x} \rangle} \mathbf{x}.$$

Compute it, and the remainder $\mathbf{r} = \mathbf{y} - \text{proj}_{\mathbf{x}}(\mathbf{y})$. Show that $\mathbf{r}$ is orthogonal to $\mathbf{x}$.

$$\text{proj}_{\mathbf{x}}(\mathbf{y}) = \frac{4}{9} \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} = \frac{1}{9} \begin{pmatrix} 4 \\ 8 \\ 8 \end{pmatrix}, \quad \mathbf{r} = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} - \frac{1}{9} \begin{pmatrix} 4 \\ 8 \\ 8 \end{pmatrix} = \frac{1}{9} \begin{pmatrix} 14 \\ -8 \\ 1 \end{pmatrix}.$$

Then $\langle \mathbf{r}, \mathbf{x} \rangle = \frac{1}{9}(14 - 16 + 2) = 0$, so $\mathbf{r}$ is orthogonal to $\mathbf{x}$. We have split $\mathbf{y}$ into a part along $\mathbf{x}$ and a part at right angles to it.

[5 marks]

(d) Not every formula that looks like an inner product is one. For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$ consider

$$\langle \mathbf{u}, \mathbf{v} \rangle = u_1 v_1 - u_2 v_2.$$

Find a vector $\mathbf{u}$ with $\langle \mathbf{u}, \mathbf{u} \rangle < 0$, and a non-zero vector with $\langle \mathbf{u}, \mathbf{u} \rangle = 0$. Which of the five properties fail? Why does this stop us defining a norm $\|\mathbf{u}\| = \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle}$?

For $\mathbf{u} = (0, 1)^\top$, $\langle \mathbf{u}, \mathbf{u} \rangle = 0 - 1 = -1 < 0$, so property 1 fails. For $\mathbf{u} = (1, 1)^\top$, $\langle \mathbf{u}, \mathbf{u} \rangle = 1 - 1 = 0$ although $\mathbf{u} \neq \mathbf{0}$, so property 2 fails. Properties 3, 4 and 5 still hold, since the formula is linear in each argument and symmetric.

The “norm” would be the square root of a negative number for $(0, 1)^\top$, and the non-zero vector $(1, 1)^\top$ would have zero length. Properties 1 and 2 are exactly what make $\sqrt{\langle \mathbf{u}, \mathbf{u} \rangle}$ a sensible length.

[6 marks]

End of question 1

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2 This question uses the properties of an inner product to prove two results that hold in every inner-product space.

(a) Consider the quadratic

$$q(t) = \langle x + ty, x + ty \rangle.$$

Use properties 3, 4 and 5 to expand $q(t)$ in the form $q(t) = At^2 + 2Bt + C$, and write down A , B and C as inner products.

Expanding using linearity in each argument (properties 3 and 4, with symmetry, property 5, to apply them to the first argument)

$$\begin{aligned} q(t) &= \langle x, x \rangle + t\langle x, y \rangle + t\langle y, x \rangle + t^2\langle y, y \rangle \\ &= \langle y, y \rangle t^2 + 2\langle x, y \rangle t + \langle x, x \rangle \end{aligned}$$

So that $A = \langle y, y \rangle$, $B = \langle x, y \rangle$ and $C = \langle x, x \rangle$.

[4 marks]

(b) By properties 1 and 2, $q(t)$ is never negative. A quadratic that is never negative cannot have two distinct real roots (it would be negative between them), so its roots are either complex or a double root. Use the formula for the roots of $At^2 + 2Bt + C = 0$ to show that this implies

$$\langle x, y \rangle^2 \leq \langle x, x \rangle \langle y, y \rangle.$$

This is the famous Cauchy-Schwarz inequality written in a very general form.

If $y = 0$ both sides of the inequality are zero, so we can take $A > 0$. The roots of $At^2 + 2Bt + C = 0$ are

$$t = \frac{-2B \pm \sqrt{4B^2 - 4AC}}{2A} = \frac{-B \pm \sqrt{B^2 - AC}}{A}.$$

The roots are complex when $B^2 - AC < 0$, and there is a double root when $B^2 - AC = 0$. If $B^2 - AC > 0$ there would be two distinct real roots and $q(t)$ would be negative between them. Thus $B^2 \leq AC$. Putting in the values of A , B and C then

$$\langle x, y \rangle^2 \leq \langle x, x \rangle \langle y, y \rangle.$$

[6 marks]

- (c) From an inner product we can define a norm $\|x\| = \sqrt{\langle x, x \rangle}$. This clearly satisfies non-negativity and “linearity” ($\|ax\| = |a|\|x\|$). The only non-trivial property to show is that this norm satisfies the triangular inequality. Expand out $\|x + y\|^2$ and hence show that

$$\|x + y\|^2 = \|x\|^2 + \|y\|^2 + 2\langle x, y \rangle$$

Then use the Cauchy-Schwarz inequality to prove the triangular inequality.

We start from

$$\begin{aligned}\|x + y\|^2 &= \langle x + y, x + y \rangle \\ &= \langle x, x \rangle + \langle y, y \rangle + 2\langle x, y \rangle\end{aligned}$$

Using $\langle x, y \rangle \leq |\langle x, y \rangle|$ and the Cauchy-Schwarz inequality $|\langle x, y \rangle|^2 \leq \langle x, x \rangle \langle y, y \rangle$ so that

$$\begin{aligned}\|x + y\|^2 &\leq \langle x, x \rangle + \langle y, y \rangle + 2\sqrt{\langle x, x \rangle \langle y, y \rangle} \\ &= \|x\|^2 + \|y\|^2 + 2\|x\|\|y\| = (\|x\| + \|y\|)^2\end{aligned}$$

Taking the square root (which is monotonic so does not change the inequality)

$$\|x + y\| \leq \|x\| + \|y\|$$

Thus proving the triangular inequality for $\|x\| = \sqrt{\langle x, x \rangle}$.

[10 marks]

End of question 2

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3 Random variables, X , Y , etc. form a vector space (i.e. they satisfy properties such as closure under addition and scalar multiplication). Furthermore we can define an inner product between random variables as

$$\langle X, Y \rangle = \mathbb{E}[XY]$$

(i.e. the expectation of the random variable $Z = XY$)

(a) Use the Cauchy-Schwarz inequality to show that

$$\text{Cov}(X, Y)^2 \leq \text{Var}(X)\text{Var}(Y)$$

hence show that the Pearson correlation is between -1 and 1 .

Let $\mu = \mathbb{E}[X]$ and $\nu = \mathbb{E}[Y]$ then

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu)(Y - \nu)]$$

From the Cauchy-Schwarz inequality

$$\begin{aligned} \text{Cov}(X, Y)^2 &= \mathbb{E}[(X - \mu)(Y - \nu)]^2 = \langle X - \mu, Y - \nu \rangle^2 \\ &\leq \langle X - \mu, X - \mu \rangle \langle Y - \nu, Y - \nu \rangle = \mathbb{E}[(X - \mu)^2] \mathbb{E}[(Y - \nu)^2] = \text{Var}(X)\text{Var}(Y). \end{aligned}$$

which proves the inequality. The Pearson correlation is defined as

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}$$

Thus $\rho(X, Y)^2 = \text{Cov}(X, Y)^2 / (\text{Var}(X)\text{Var}(Y))$, but we have already shown that $\text{Cov}(X, Y)^2 \leq \text{Var}(X)\text{Var}(Y)$ so $\rho(X, Y)^2 \leq 1$ and $\rho(X, Y)$ will lie in the interval $[-1, 1]$.

[10 marks]

(b) Show that for vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ that the inner product

$$\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^n w_i x_i y_i$$

satisfies the conditions of an inner product provided $w_i > 0$ for all i . Write down the norm and distance induced by this inner product and provide an interpretation of what this distance means.

We consider the five conditions on the inner product

(i) Non-negativity follows from

$$\langle \mathbf{x}, \mathbf{x} \rangle = \sum_{i=1}^n w_i x_i^2 \geq 0$$

since for each term in the sum $w_i x_i^2 \geq 0$.

(ii) For $\langle \mathbf{x}, \mathbf{x} \rangle = 0$ each term in the sum must be zero. Since $w_i > 0$ this implies $x_i = 0$ for all i , that is, $\mathbf{x} = \mathbf{0}$.

(iii) Linearity follows trivially since

$$\langle a\mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^n w_i (ax_i) y_i = a \sum_{i=1}^n w_i x_i y_i = a \langle \mathbf{x}, \mathbf{y} \rangle$$

(iv) The distributive law again follows trivially

$$\begin{aligned} \langle \mathbf{x}, \mathbf{y} + \mathbf{z} \rangle &= \sum_{i=1}^n w_i x_i (y_i + z_i) \\ &= \sum_{i=1}^n w_i x_i y_i + \sum_{i=1}^n w_i x_i z_i = \langle \mathbf{x}, \mathbf{y} \rangle + \langle \mathbf{x}, \mathbf{z} \rangle \end{aligned}$$

(v) Finally symmetry follows from the commutativity of multiplication

$$\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^n w_i x_i y_i = \sum_{i=1}^n w_i y_i x_i = \langle \mathbf{y}, \mathbf{x} \rangle$$

The induced norm and distance for this inner product are

$$\|\mathbf{x}\| = \sqrt{\sum_{i=1}^n w_i x_i^2}, \quad d(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{i=1}^n w_i (x_i - y_i)^2}.$$

A natural interpretation of this is that this is a generalisation of the Euclidean distance where we rescale each axis by a factor $\sqrt{w_i}$ and then compute the normal Euclidean distance. A large w_i means that differences in feature i count for a lot.

[10 marks]

End of question 3

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Answers

4 In the previous question you showed that $\langle x, y \rangle = \sum_{i=1}^n w_i x_i y_i$ is an inner product provided every weight w_i is positive. Functions form a vector space in just the same way (we can add them and multiply them by scalars), and if we replace the sum over components by an integral over x we get

$$\langle f, g \rangle = \int_{-\infty}^{\infty} w(x) f(x) g(x) dx$$

which, by the same argument, is an inner product provided the *weight function* $w(x)$ is positive.

In this question we choose $w(x)$ to be the density of a standard normal deviate

$$w(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

so that $\langle f, g \rangle = \mathbb{E}[f(X)g(X)]$ where $X \sim \mathcal{N}(0,1)$. That is, this is exactly the inner product of the previous question, applied to functions of a random variable.

The monomials $f_i(x) = x^i$ span the polynomials, but they are not orthogonal. We can orthogonalise them with the Gram–Schmidt procedure: having found $p_0, p_1, \dots, p_{n-1}$ we subtract from f_n its component along each of them,

$$p_n(x) = f_n(x) - \sum_{i=0}^{n-1} \frac{\langle f_n, p_i \rangle}{\langle p_i, p_i \rangle} p_i(x), \quad p_0(x) = 1.$$

Orthogonal polynomials are useful because they make approximating a function easy. If we want the best polynomial approximation to some function f we can compute each coefficient separately with a single inner product, and if we later decide to add a higher order term none of the coefficients we have already found change. Using the monomials instead we would have to solve a set of simultaneous equations, and start again from scratch each time we added a term.

You will need the integrals below, which you are not expected to compute. (They are the moments of a standard normal deviate. All the odd moments vanish because $w(x)x^n$ is an odd function.)

n	0	1	2	3	4	5	6
$\mathbb{E}[X^n] = \int_{-\infty}^{\infty} w(x) x^n dx$	1	0	1	0	3	0	15

- (a) Show that $\langle f, g \rangle = \int w(x) f(x) g(x) dx$ satisfies the five conditions for an inner product given at the start of this sheet, provided $w(x) > 0$.

The argument is the same as for the weighted sum, with the sum over i replaced by the integral over x .

- (i) $\langle f, f \rangle = \int w(x) f^2(x) dx \geq 0$, since the integrand $w(x) f^2(x) \geq 0$ everywhere.
 - (ii) If $\langle f, f \rangle = 0$ then, as the integrand is non-negative and $w(x) > 0$, we need $f(x) = 0$ (for all but a set of points of zero total length).
 - (iii) $\langle af, g \rangle = \int w(x) af(x) g(x) dx = a \langle f, g \rangle$.
 - (iv) $\langle f, g + h \rangle = \int w(x) f(x) (g(x) + h(x)) dx = \langle f, g \rangle + \langle f, h \rangle$, as the integral of a sum is the sum of the integrals.
 - (v) $\langle f, g \rangle = \langle g, f \rangle$ since $f(x)g(x) = g(x)f(x)$.
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[5 marks]

- (b) Starting from $p_0(x) = 1$, use Gram–Schmidt to find $p_1(x)$ and $p_2(x)$.

For p_1 we subtract from $f_1(x) = x$ its component along p_0 . Since $\langle f_1, p_0 \rangle = \mathbb{E}[X] = 0$,

$$p_1(x) = x - \frac{0}{1} 1 = x.$$

For p_2 we start from $f_2(x) = x^2$ and use $\langle p_0, p_0 \rangle = \mathbb{E}[1] = 1$ and $\langle p_1, p_1 \rangle = \mathbb{E}[X^2] = 1$,

$$\langle f_2, p_0 \rangle = \mathbb{E}[X^2] = 1, \quad \langle f_2, p_1 \rangle = \mathbb{E}[X^3] = 0,$$

so that

$$p_2(x) = x^2 - \frac{1}{1} 1 - \frac{0}{1} x = x^2 - 1.$$

[5 marks]

(c) Find $p_3(x)$.

We need $\langle p_2, p_2 \rangle = \mathbb{E}[(X^2 - 1)^2] = \mathbb{E}[X^4] - 2\mathbb{E}[X^2] + 1 = 3 - 2 + 1 = 2$, and starting from $f_3(x) = x^3$

$$\begin{aligned}\langle f_3, p_0 \rangle &= \mathbb{E}[X^3] = 0, & \langle f_3, p_1 \rangle &= \mathbb{E}[X^4] = 3, \\ \langle f_3, p_2 \rangle &= \mathbb{E}[X^3(X^2 - 1)] = \mathbb{E}[X^5] - \mathbb{E}[X^3] = 0.\end{aligned}$$

Therefore

$$p_3(x) = x^3 - \frac{0}{1}1 - \frac{3}{1}x - \frac{0}{2}(x^2 - 1) = x^3 - 3x.$$

(These are the (probabilists') Hermite polynomials. As a check, every p_n is either an odd or an even function, which had to happen because $w(x)$ is symmetric about the origin.)

[5 marks]

(d) If we write a function as $f(x) = \sum_i a_i p_i(x)$, show that $a_i = \langle f, p_i \rangle / \langle p_i, p_i \rangle$. Use this to write $f(x) = x^3$ in terms of p_0, p_1, p_2 and p_3 , and check your answer.

Taking the inner product of $f = \sum_j a_j p_j$ with p_i

$$\langle f, p_i \rangle = \sum_j a_j \langle p_j, p_i \rangle = a_i \langle p_i, p_i \rangle$$

because $\langle p_j, p_i \rangle = 0$ unless $i = j$. Rearranging gives $a_i = \langle f, p_i \rangle / \langle p_i, p_i \rangle$. This is the point of orthogonality: one integral gives one coefficient.

For $f(x) = x^3$ we computed the three inner products we need in the previous part, and

$$\langle f, p_3 \rangle = \mathbb{E}[X^3(X^3 - 3X)] = \mathbb{E}[X^6] - 3\mathbb{E}[X^4] = 15 - 9 = 6 \text{ with}$$

$$\langle p_3, p_3 \rangle = \mathbb{E}[(X^3 - 3X)^2] = \mathbb{E}[X^6] - 6\mathbb{E}[X^4] + 9\mathbb{E}[X^2] = 15 - 18 + 9 = 6.$$

Hence $a_0 = 0$, $a_1 = 3/1 = 3$, $a_2 = 0$ and $a_3 = 6/6 = 1$, giving

$$x^3 = 3p_1(x) + p_3(x).$$

Checking, $3x + (x^3 - 3x) = x^3$ as required.

[5 marks]

End of question 4

END OF PAPER